

ROBOTICS ASSIGNMENT

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The first part of the assignment concerns the trajectory planning for the door handle. For its design, the following procedure was followed.

Initially, the overall motion was modeled as a combination of two independent angular trajectories: angle θ_1 , which describes the rotation of the door, and angle θ_2 , which describes the rotation of the handle. In addition, the motion was divided into three successive time phases within the time interval 0–5 seconds.

The time phases are as follows:

1. **Phase 1 (0–1.5 sec): Rotation of the handle from $\theta_2=0^\circ$ to $\theta_2=-45^\circ$**
2. **Phase 2 (1.5–3.5 sec): Rotation of the door from $\theta_1=0^\circ$ to $\theta_1=-30^\circ$, with the handle fixed.**
3. **Phase 3 (3.5–5 sec): Return of the handle from $\theta_2=-45^\circ$ to $\theta_2=0^\circ$, with the door fixed.**

The angular trajectories θ_1 and θ_2 are calculated using a 5th-degree polynomial with zero initial and final velocities and accelerations. The function used has the form:

$q(t)=q_0+(q_f-q_0)\cdot(10t^3-15t^4+6t^5)$ where q_0 and q_f are the initial and final values of the angles, and t_0 to t_f is the time duration of the phase.

Calculation of homogeneous transformations of the door and the handle:

The door is considered to rotate around the vertical axis Z by an angle $\theta_1(t)$. The orientation of the door frame with respect to the fixed frame $\{0\}$ is given at each instant by the rotation matrix:

$$R_{OD}(t) = \text{Rot}_z(\theta_1(t))$$

The complete homogeneous transformation matrix of the door is:

$$G_{OD}(t) = \text{transl}(0, 2, 0) \cdot r_{2t}(R_{OD})$$

The handle frame is calculated with respect to the door frame through a sequence of transformations. Initially, a translation of 0.9 along the x-axis and 0.7 along the z-axis is applied in order to place the handle at the correct height and distance on the surface of the door. Next, we calculate the rotation matrix R:

$$R = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Finally, a rotation around the x_h axis of the handle is applied using the matrix:

$$R_{knob}(t) = \text{Rot}_x(\theta_2(t))$$

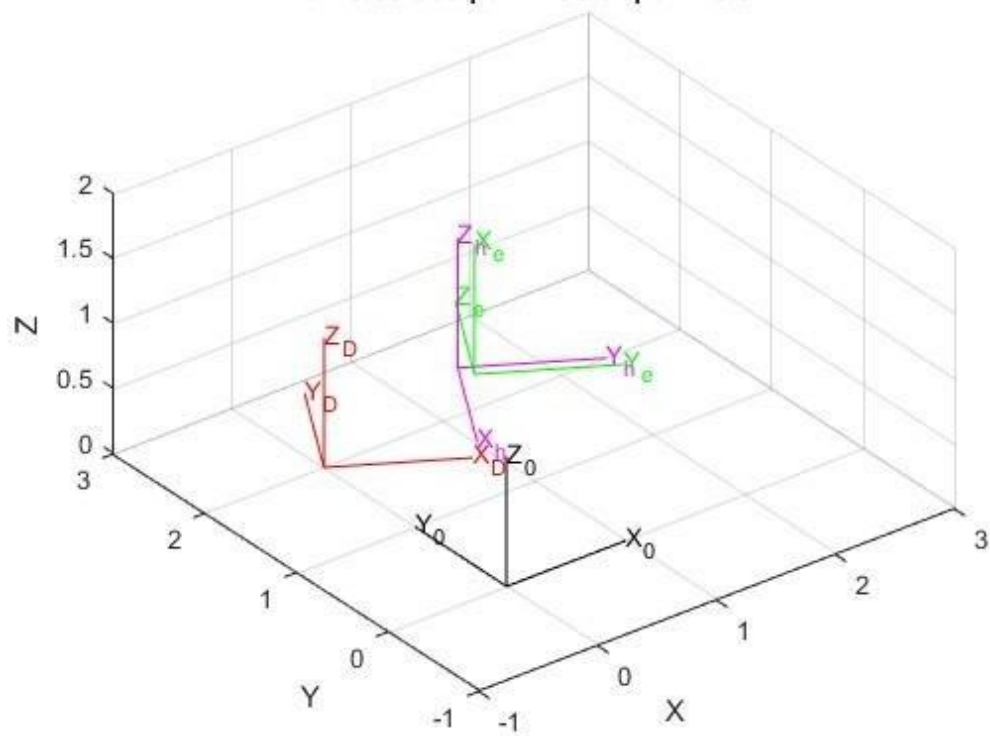
The overall transformation of the handle with respect to frame $\{0\}$:

$$G_{DH}(t) = \text{transl}(0.9, 0, 0.7) * r_{2t}(R_h) * r_{2t}(R_{knob})$$

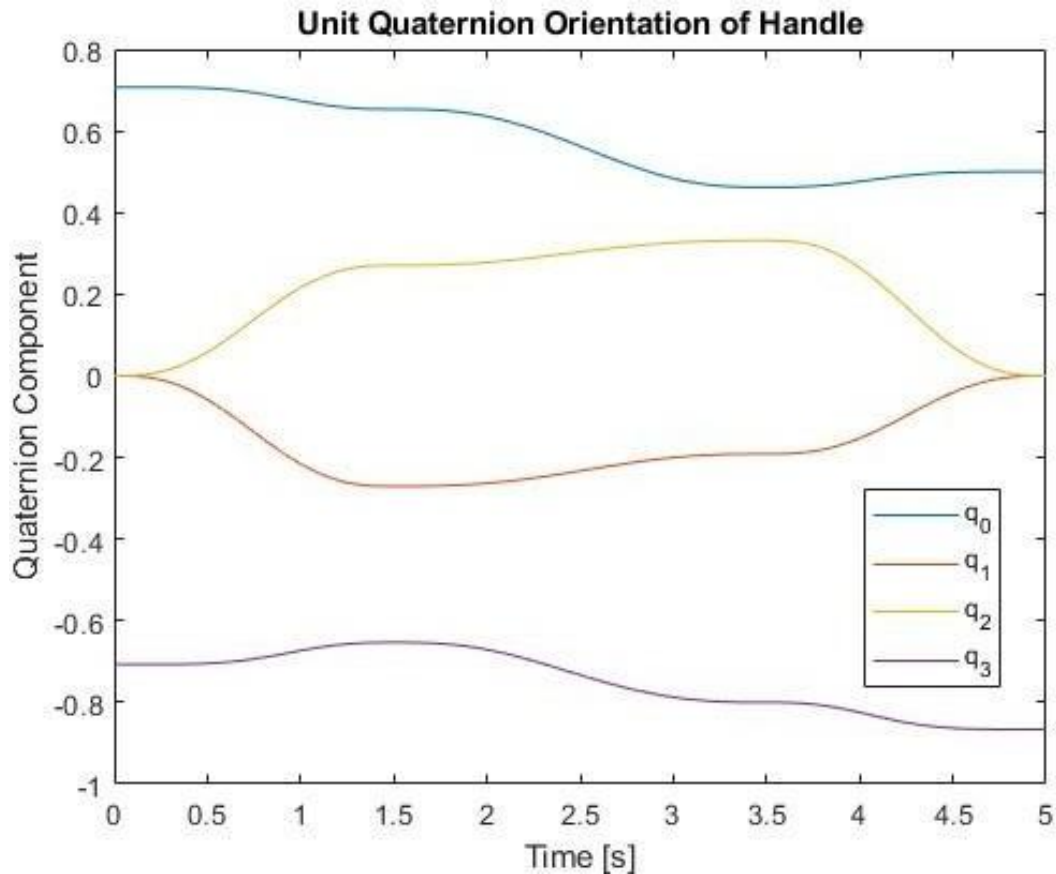
And finally, the homogeneous transformation G_{OH} is calculated from the product: $G_{OH}(t) = G_{OD} * G_{DH}$

Final simulation position

$t = 5.00 \text{ sec} \mid \theta_1 = -30.0^\circ \mid \theta_2 = 0.0^\circ$



Orientation using Unit Quaternion:



The diagram shown concerns the trajectory of the handle orientation, expressed through the components of a Unit Quaternion (q_0, q_1, q_2, q_3) as a function of time. From the curve, we can observe that all components change smoothly and continuously during the motion from 0 to 5 seconds, which confirms a smooth trajectory without discontinuities or sudden jumps.

The time phases of the motion are distinguished as follows: In the first stage (0–1.5 s), the handle rotates from the initial position to the maximum rotation, with components q_1 and q_2 showing a characteristic increase and decrease respectively. In the second phase (1.5–3.5 s), the handle remains in a fixed position, which is reflected in the almost constant value of the quaternions. Finally, during the third phase (3.5–5 s), the handle returns to its initial position, with the components returning symmetrically to their initial values, which confirms the correct trajectory of the handle.

At the same time, q_0 starts from a high value and decreases slightly as the handle rotates, indicating that a smooth rotation takes place

without abrupt changes, while q_3 remains constantly negative and changes very little, which means that the rotation takes place steadily around the vertical z -axis.

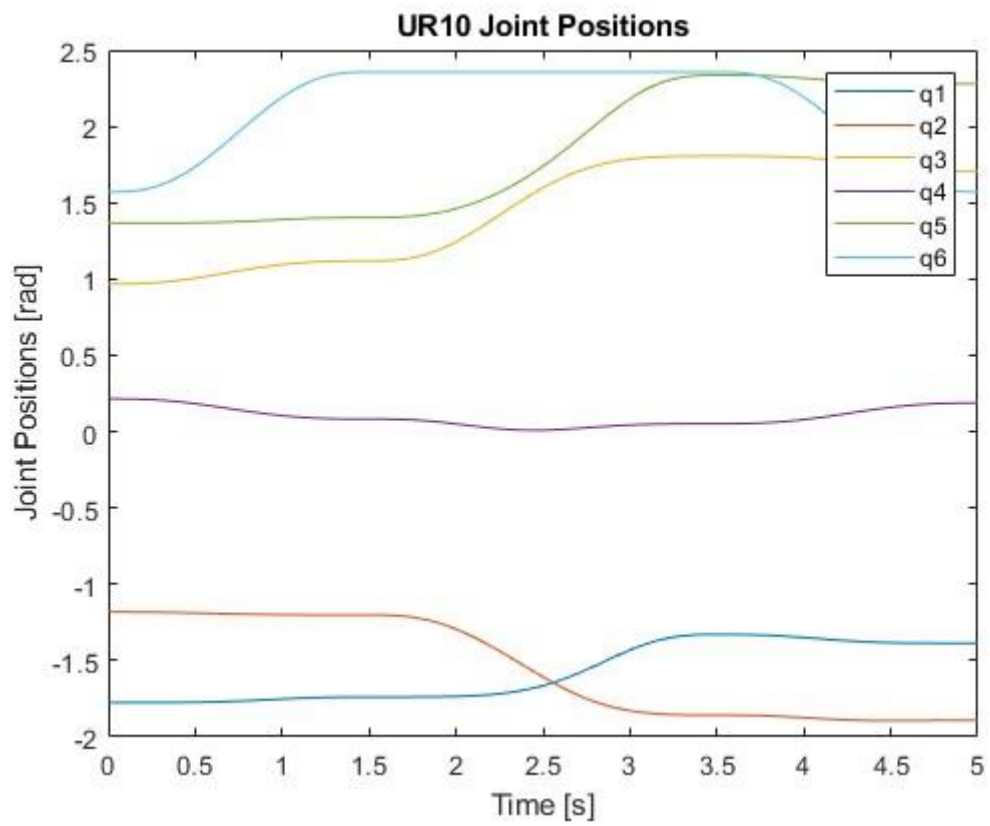
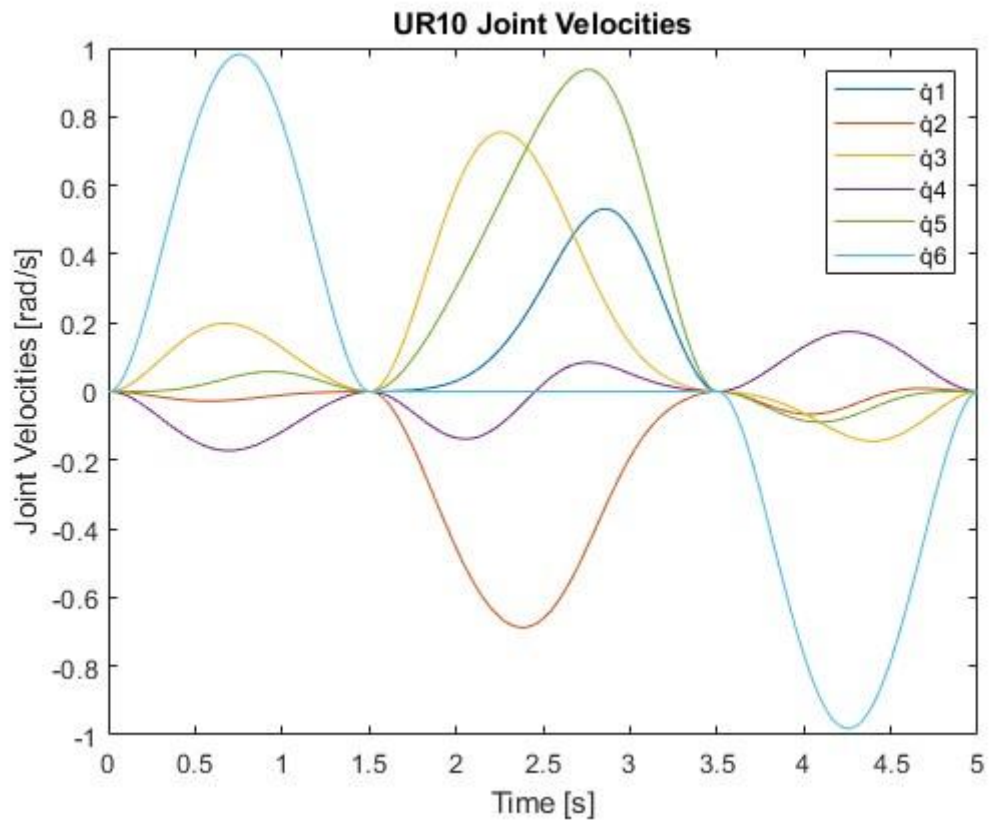
Overall, the diagram confirms the correctness of the implementation of the handle orientation trajectory.

Regarding the second part of the assignment, the end-effector frame $\{e\}$ was defined in order to calculate and store its trajectory during the motion. This was achieved by calculating the homogeneous transformation between the handle frame $\{h\}$ and the tool frame $\{e\}$. Specifically, the orientation of frame $\{e\}$ with respect to $\{h\}$ is calculated as:

$${}^hR_e = \begin{bmatrix} 0 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

, while $PHE = \text{transl}(0.1, 0.1, 0)$ defines the relative position of the tool with respect to the handle. The total homogeneous transformation $G_{0E} = \text{transl}(0.1, 0.1, 0) * r2t(R_{EH})$ expresses the position and orientation of the tool with respect to the handle. Then, through the product $G_{0E} = G_{0H} * G_{EH}$, the position and orientation of the tool with respect to the initial frame $\{0\}$ are calculated for each instant of time.

Next, through G_{0E} , the linear and angular velocities of the end-effector are calculated at each instant, with the help of the `tr2delta` function. Then, the end-effector Jacobian of the UR10 for the given joint position is calculated using the command `J = ur10.jacobe(q)`. Thus, the joint velocities are calculated as the product of the inverse Jacobian with the velocity vector of the tool end-effector. From the simulation, a tool position error is observed at the final position $e_x = x_d - x = -0.0011$, $e_y = y_d - y = 0.0007$, $e_z = z_d - z = 0.0010$, where x_d, y_d, z_d are the desired coordinates and x, y, z are the final actual coordinates with respect to frame $\{0\}$.



From the diagram of the joint positions (q_1 to q_6), we observe that the joints follow smooth and continuous trajectories without abrupt changes throughout the 5 seconds. Each joint participates differently in the motion, with some showing a more pronounced variation (e.g. q_1 and q_5), while others (e.g. q_4) remain relatively stable.

Similarly, in the joint velocity diagram, it is found that all velocities are continuous and smooth functions of time, without discontinuities or peaks. The value of q_6 correctly remains 0 in the interval 1.5–3.5 sec, since in this phase the handle does not rotate while the door opens, whereas in the other two phases it shows an increase and decrease respectively as the handle opens and closes.